Assignment-2

Objective

Analyze a retail business dataset using SQL. The database contains customers, products, employees, orders, order line items, and returns. Students are provided with the data and are expected to write SQL queries to answer business questions. Database/table creation and data entry are not part of the assessed tasks.

Tables

- **Customers:** Customer details, location, and signup date.
- **Products:** Product category, current price, and stock quantity.
- **Employees:** Sales/support/operations staff, branch city, and monthly salary.
- **Orders:** Order dates, customers, assigned employees, status, and payment method.
- **Order_Items:** Products, quantities, sale-time prices, and discount rates per order.
- **Returns:** Return dates and reasons for returned orders.

Use the provided Excel workbook, CSV files, or SQL setup script supplied with this project. The tables have already been designed and populated.

Business Rules / Assumptions

- Line-item gross value = quantity × unit_price_at_sale.
- Line-item net value after discount = quantity × unit_price_at_sale × (1 − discount_rate).
- Unless a question says otherwise, revenue calculations should include only orders with order_status = 'Delivered'.
- A returned order is recorded in Returns and normally has order_status = 'Returned'. Do not count returned or cancelled orders as delivered revenue.

Queries to Perform

Q1. Basic filtering and sorting (3 marks)

List delivered orders placed between 1 October 2024 and 31 March 2025, showing order_id, order_date, customer_id, and payment_method. Sort by order_date descending.

Q2. Multi-table JOIN (4 marks)

Display each delivered order with customer_name, order_id, order_date, employee_name, and order_status. Include only orders with matching customer and employee records.

Q3. Aggregation and HAVING (4 marks)

For each product category, calculate total units sold and net revenue from delivered orders. Show only categories with net revenue greater than ₹10,000; sort by net revenue descending.

Q4. Customers with repeat purchases (3 marks)

Find customers who have placed at least two delivered orders. Show customer_id, customer_name, number of delivered orders, and total net spend.

Q5. LEFT JOIN and zero activity (3 marks)

List all customers and the number of delivered orders each has placed, including customers with zero delivered orders.

Q6. Subquery (4 marks)

Find products whose unit_price is greater than the average unit_price of all products. Display product name, category, unit price, and the overall average price.

Q7. NOT EXISTS (4 marks)

Find customers who have never placed a delivered order. Return their customer_id, customer_name, and city.

Q8. Date functions (3 marks)

Group delivered orders by year and month, showing the number of orders and net revenue for each month. Sort chronologically.

Q9. CASE expression (3 marks)

Classify each product as 'Low Stock' when stock_quantity < 20, 'Moderate Stock' when stock_quantity is between 20 and 40 inclusive, and 'High Stock' when stock_quantity > 40. Display product_name, stock_quantity, and stock label.

Q10. Conditional aggregation (4 marks)

For each employee, calculate delivered, returned, and cancelled order counts using conditional aggregation. Include employees with no orders.

Q11. Window function – ranking (4 marks)

Rank products within each category by net revenue from delivered orders, highest first. Show category, product_name, net revenue, and rank. Products with no delivered sales should still be eligible to appear with zero revenue.

Q12. Running total (4 marks)

For each month, calculate delivered-order net revenue and a cumulative running total of revenue across months in chronological order.

Q13. Return analysis (3 marks)

List each returned order with customer name, return date, return reason, and the total pre-discount value of its order items.

Q14. CTE (4 marks)

Using a CTE, calculate net revenue per customer from delivered orders, then return customers whose net revenue is above the average customer net revenue. State whether the average is calculated across all customers or only customers with delivered purchases.

Q15. Business insight query (4 marks)

Identify the top three customers by delivered net spend. Show customer name, number of delivered orders, net spend, and each customer's percentage contribution to total delivered net revenue.

Instructions

- **Task Completion:** Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.
- **Assumptions:** Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation.
- **GitHub Repository:** Create a GitHub repository to host your project. Upload your project files, including source code and documentation, to the repository.
- **No Copying:** Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.
- **Submission:** Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or assigned. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project!

Important Note

The project brief contains questions only. The data is provided separately in the accompanying workbook and dataset bundle.

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